

AP Calc Lesson 2.8 Homework

Answer Key

1. $f'(x) = \frac{(x^2 + 1)e^x - e^x(2x)}{(x^2 + 1)^2} = \frac{e^x(x^2 - 2x + 1)}{(x^2 + 1)^2}$ or equivalent

2. a. $f(x)$ can be simplified to $f(x) = 1 + \frac{4}{x}$
 b. $f'(x) = \frac{-4}{x^2}$ or equivalent

3. x^3

4. a. $g'(x) = \frac{f(x)\left(\frac{2}{x}\right) - (2\ln x)(f'(x))}{(f(x))^2}$
 b. $g'(e) = \frac{f(e)\left(\frac{2}{e}\right) - (2\ln e)(f'(e))}{(f(e))^2} = \frac{\frac{4}{e} - 2(6)}{4} = \frac{1 - 3e}{e}$ or equivalent

5. $h'(\pi) = \frac{(2\cos \pi)\left(-\frac{2}{\pi}\right) - (2)(-2\sin \pi)}{(2\cos \pi)^2} = \frac{-2\left(-\frac{2}{\pi}\right) - 2(0)}{(-2)^2} = \frac{\frac{4}{\pi}}{4} = \frac{1}{\pi}$

6. a. Positive. $h'(1) = \frac{(11)(+) - (3)(-2)}{(11)^2} > 0$
 b. Positive. $h'(3) = \frac{(7)(0) - (6)(-2)}{(7)^2} > 0$
 c. Cannot be determined. $h'(5) = \frac{(3)(-) - (3)(-2)}{(3)^2}$.

If $f'(5) < -2$, $h'(5)$ would be negative.

If $f'(5) = -2$, $h'(5)$ would be zero.

If $f'(5) > -2$, $h'(5)$ would be positive.

7. a. $\frac{dy}{dx} = \frac{e^x(2x) - x^2e^x}{(e^x)^2} = \frac{2x - x^2}{e^x}$
 $\frac{dy}{dx} = 0 \Rightarrow 2x - x^2 = 0 \Rightarrow x = 0, x = 2$ giving the points $(0, 0)$ and $\left(2, \frac{4}{e^2}\right)$.

b. A vertical tangent would have an undefined slope. The slope of the tangent would be undefined when the denominator is equal to zero. Since $e^x \neq 0$, the slope is never undefined and thus, the tangent to the curve is never vertical.

8. At the point $\left(10, \frac{5}{2}\right)$, $c = 50$ and at the point $\left(2, -\frac{1}{2}\right)$, $c = 2$.

9. B

10. C